

BUSINESS PROBLEM STATEMENT

A leading retail company aims to gain a deeper understanding of its customers' shopping behavior to improve sales performance, customer satisfaction, and long-term customer loyalty. The management team has observed changes in purchasing patterns across demographics, product categories, and sales channels (online and offline).

The company is particularly interested in identifying the key factors that influence consumer purchasing decisions and repeat purchases, including discounts, customer reviews, seasonal trends, and payment preferences.

You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

DELIVERABLES

1. Data Preparation & Modeling (Python)

Clean, preprocess, transform, and prepare the raw dataset using Python to ensure it is suitable for reliable and meaningful analysis.

2. Data Analysis (SQL)

Organize the data into a structured format, simulate relevant business transactions, and execute SQL queries to extract insights related to customer segments, loyalty, purchasing behavior, and key purchase drivers.

3. Visualization & Insights (Power BI)

Develop an interactive Power BI dashboard that presents key patterns, trends, customer behavior, and business insights, enabling stakeholders to make informed, data-driven decisions.

4. Report and Presentation

Prepare a clear and professional project report summarizing the key findings, insights, and business recommendations. Develop a presentation that effectively communicates the major insights and actionable recommendations to stakeholders.

5. Repository

Maintain a well-structured project repository containing all Python scripts, SQL queries, processed data files where appropriate, and Power BI/dashboard files.